

EXPERIMENT NO. 1 -BASIC CALCULATOR

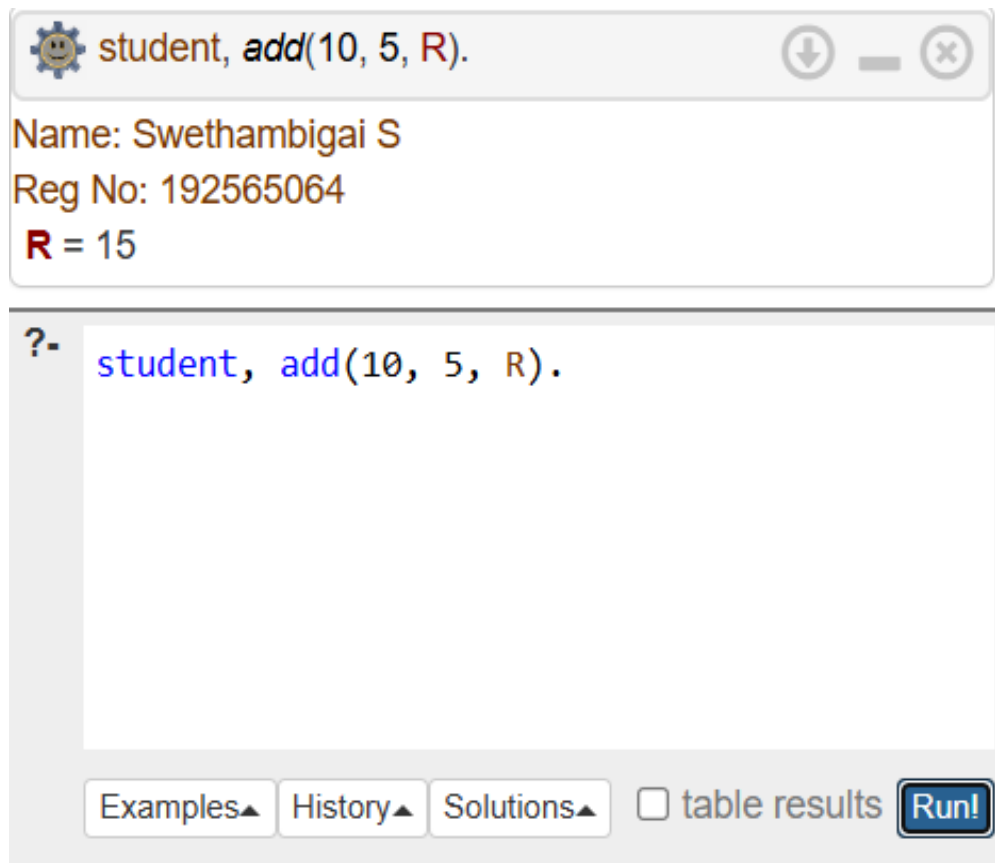
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
add(A, B, R) :-  
    R is A + B.  
  
subtract(A, B, R) :-  
    R is A - B.  
  
multiply(A, B, R) :-  
    R is A * B.  
  
divide(A, B, R) :-  
    B =\= 0,  
    R is A / B.
```

Query

```
student, add(10, 5, R).
```

Output



EXPERIMENT NO. 2-ALL COMPARISON

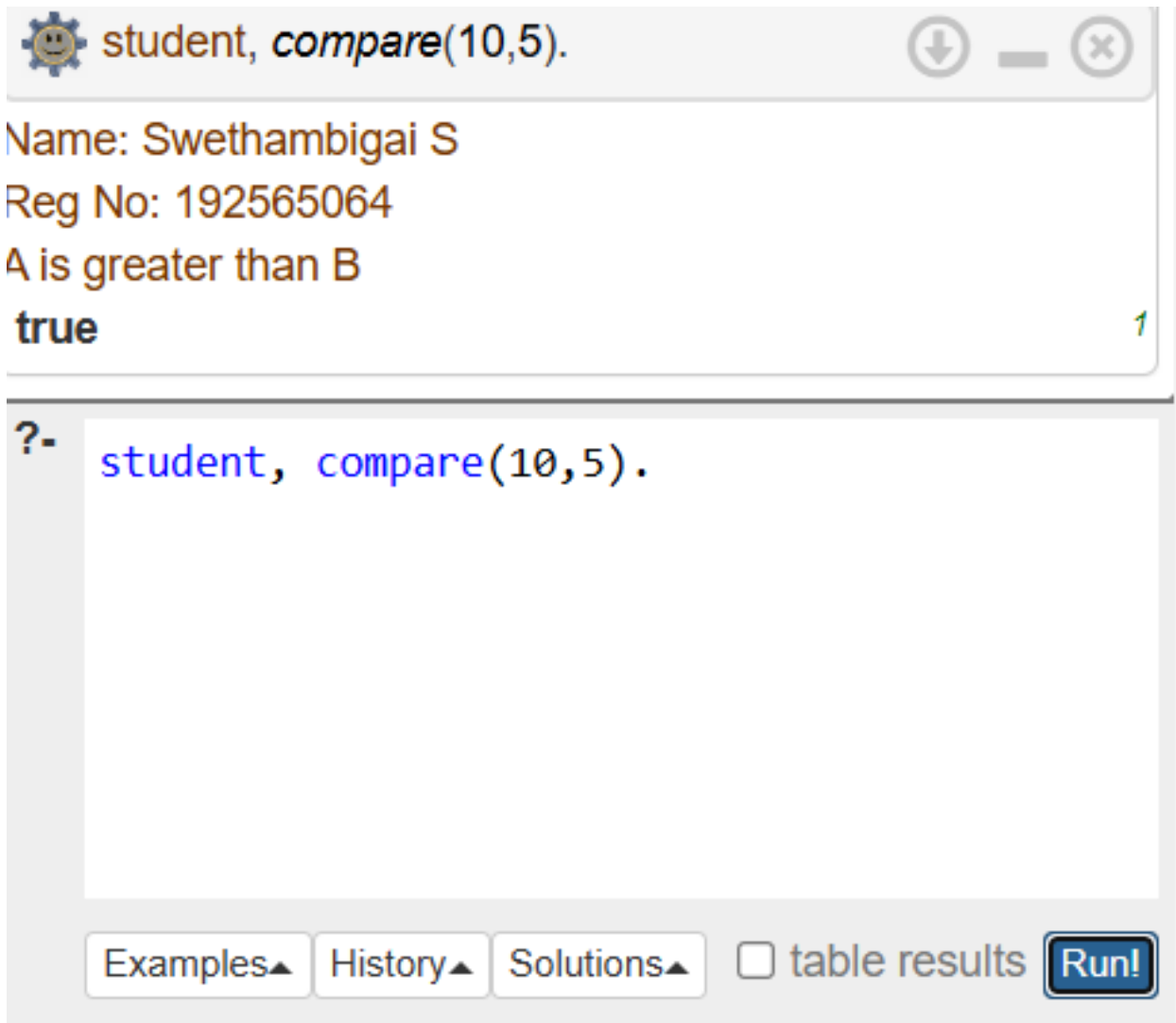
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
compare(A,B) :-  
    A > B, write('A is greater than B').
```

Query

```
student, compare(10,5).
```

Output



The screenshot shows a Prolog IDE interface. At the top, a title bar contains a gear icon, the text "student, compare(10,5).", and standard window controls (download, minimize, close). Below the title bar, the output area displays the results of the query: "Name: Swethambigai S", "Reg No: 192565064", "A is greater than B", and "true". A small green "1" is visible in the bottom right corner of the output area. Below the output area, a query input field shows "?- student, compare(10,5).". At the bottom of the interface, there are four buttons: "Examples▲", "History▲", "Solutions▲", and "table results" (with an unchecked checkbox). To the right of these buttons is a blue "Run!" button.

EXPERIMENT NO. 3-VOTE ELIGIBILITY

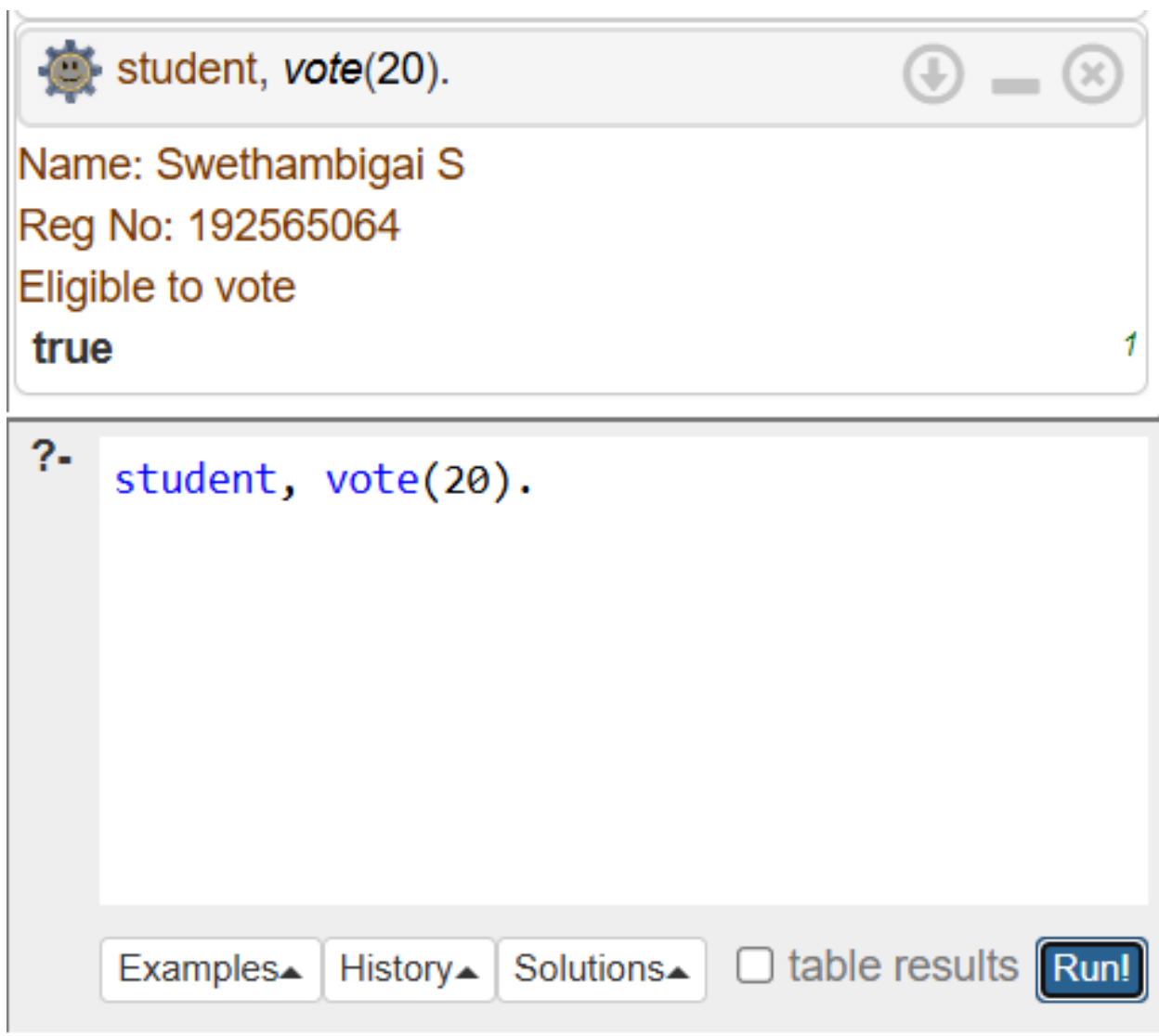
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
vote(Age) :-  
    Age >= 18,  
    write('Eligible to vote').
```

Query

```
student, vote(20).
```

Output



The screenshot shows a Prolog IDE window with a title bar containing a gear icon, the text "student, vote(20).", and standard window controls (download, minimize, close). The main area displays the output of the query: "Name: Swethambigai S", "Reg No: 192565064", "Eligible to vote", and "true". A green "1" is visible in the bottom right corner of the output area. Below the output area, there is a query input field with a "?-" prompt and the text "student, vote(20).". At the bottom of the window, there are four buttons: "Examples▲", "History▲", "Solutions▲", and "table results" (with an unchecked checkbox). To the right of these buttons is a blue "Run!" button.

EXPERIMENT NO. 4-CELSIUS TO FAHRENHEIT

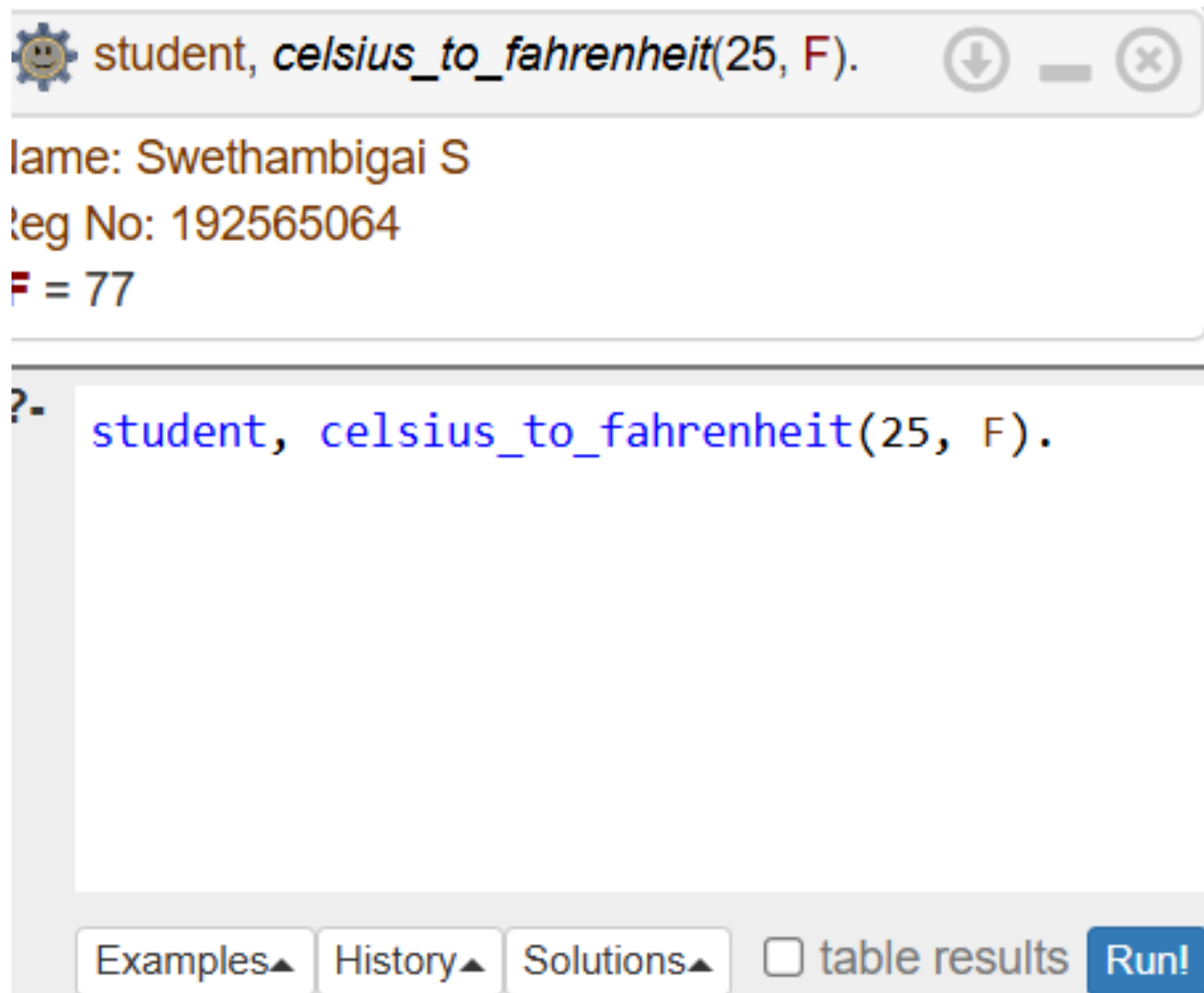
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
celsius_to_fahrenheit(C, F) :-  
    F is (C * 9 / 5) + 32.
```

Query

```
student, celsius_to_fahrenheit(25, F).
```

Output



The screenshot shows a Prolog IDE window with a title bar containing a gear icon, a smiley face icon, and window controls. The main area displays the output of the query `student, celsius_to_fahrenheit(25, F).` as follows:

```
Name: Swethambigai S  
Reg No: 192565064  
F = 77
```

Below the output, there is a prompt `?-` followed by the query `student, celsius_to_fahrenheit(25, F).` in a text input area.

At the bottom of the window, there are four buttons: "Examples▲", "History▲", "Solutions▲", and "table results" (with a checkbox icon). To the right of these buttons is a blue "Run!" button.

EXPERIMENT NO. 5-PASS / FAIL - 3 SUBJECTS

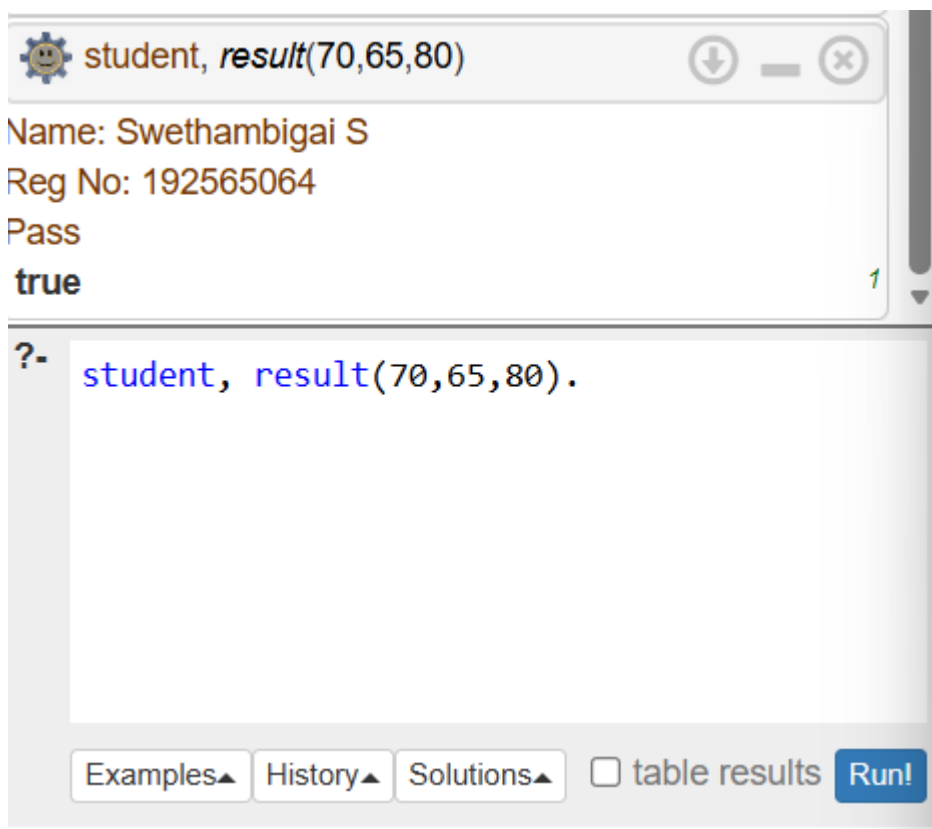
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
result(A,B,C) :-  
    A >= 40, B >= 40, C >= 40,  
    write('Pass').
```

Query

```
student, result(70,65,80).
```

Output



The screenshot shows a Prolog IDE window titled "student, result(70,65,80)". The output area displays the results of the query: "Name: Swethambigai S", "Reg No: 192565064", "Pass", and "true". A green "1" is visible in the bottom right corner of the output area. The query input area shows "?- student, result(70,65,80).". At the bottom, there are buttons for "Examples", "History", "Solutions", a checkbox for "table results", and a "Run!" button.

```
student, result(70,65,80)
```

Name: Swethambigai S
Reg No: 192565064
Pass
true

?- student, result(70,65,80).

Examples History Solutions ☐ table results Run!

EXPERIMENT NO. 6- AVERAGE OF 5 NUMBERS

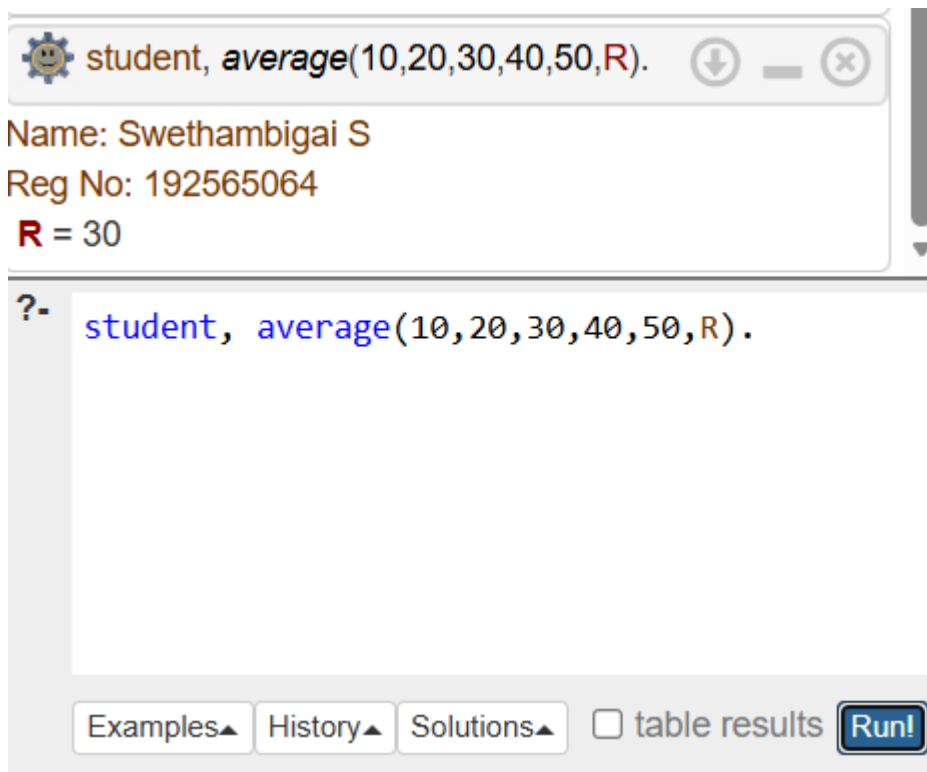
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
average(A,B,C,D,E,R) :-  
    R is (A+B+C+D+E)/5.
```

Query

```
student, average(10,20,30,40,50,R).
```

Output



The screenshot shows a Prolog IDE window. The title bar contains a gear icon, the text "student, average(10,20,30,40,50,R).", and standard window controls. The main area displays the output of the query: "Name: Swethambigai S", "Reg No: 192565064", and "R = 30". Below the output, there is a prompt "?-" followed by the query "student, average(10,20,30,40,50,R).". At the bottom, there are buttons for "Examples", "History", "Solutions", a checkbox for "table results", and a "Run!" button.

```
student, average(10,20,30,40,50,R).
```

Name: Swethambigai S
Reg No: 192565064
R = 30

?- student, average(10,20,30,40,50,R).

Examples History Solutions ☐ table results Run!

EXPERIMENT NO. 7-MULTIPLICATION TABLE

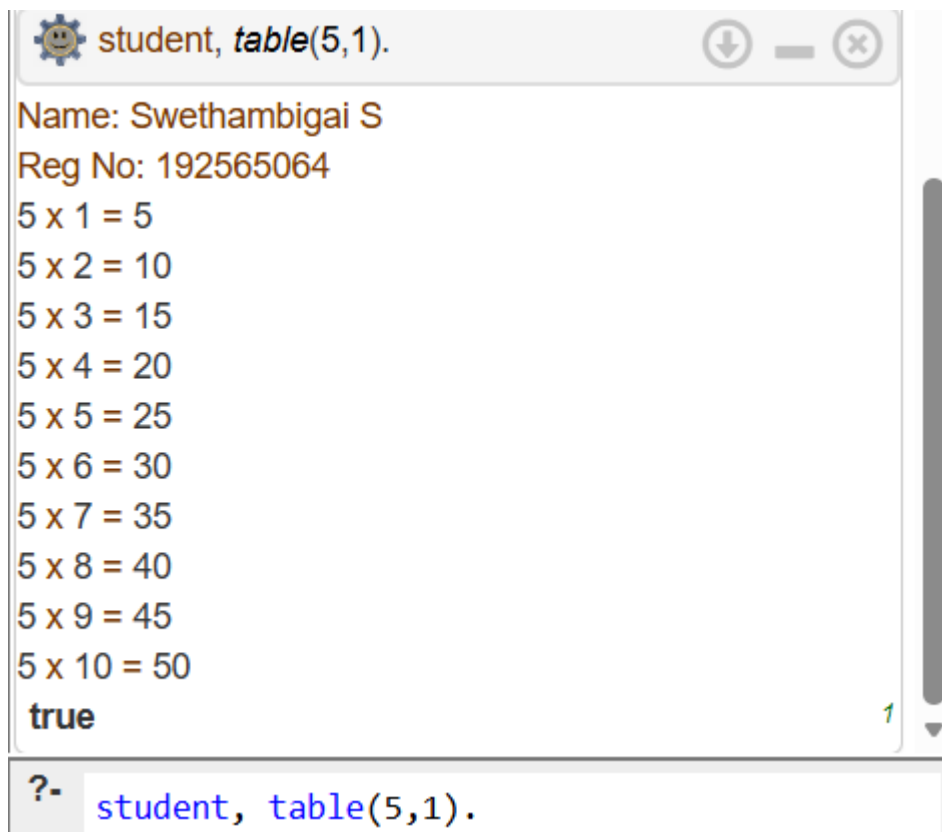
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
table(N,I) :-  
    I =< 10,  
    R is N*I,  
    write(N), write(' x '), write(I), write(' = '), write(R), nl,  
    I1 is I+1,  
    table(N,I1).  
table(_,11).
```

Query

```
student, table(5,1).
```

Output



```
student, table(5,1).  
Name: Swethambigai S  
Reg No: 192565064  
5 x 1 = 5  
5 x 2 = 10  
5 x 3 = 15  
5 x 4 = 20  
5 x 5 = 25  
5 x 6 = 30  
5 x 7 = 35  
5 x 8 = 40  
5 x 9 = 45  
5 x 10 = 50  
true  
?- student, table(5,1).
```

EXPERIMENT NO. 8

STUDENT TEACHER SUBJECT RELATION

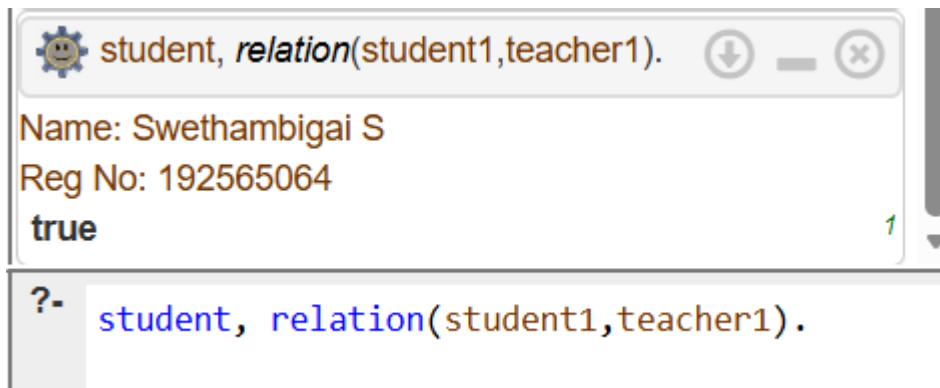
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
teaches(teacher1,prolog).  
studies(student1,prolog).  
  
relation(S,T) :-  
    studies(S,prolog),  
    teaches(T,prolog).
```

Query

```
student, relation(student1,teacher1).
```

Output



EXPERIMENT NO. 9

FAMILY TREE – ALL RELATIONS

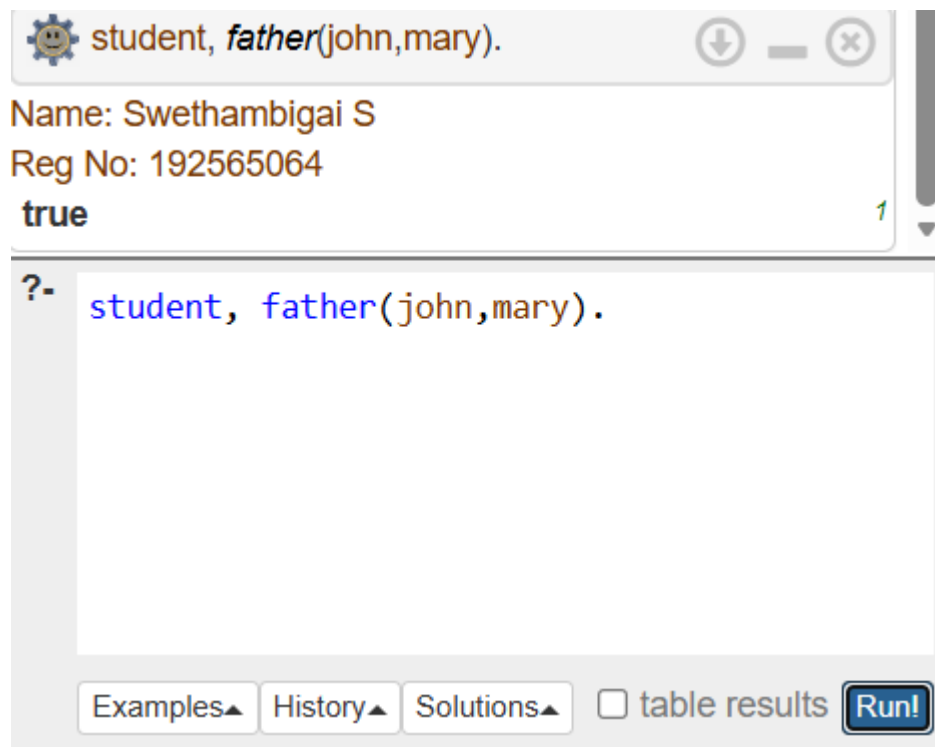
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
male(john).  
female(mary).  
parent(john,mary).  
  
father(X,Y) :-  
    male(X), parent(X,Y).  
  
mother(X,Y) :-  
    female(X), parent(X,Y).
```

Query

```
student, father(john,mary).
```

Output



The screenshot shows a Prolog IDE window with a title bar containing a gear icon, the text "student, father(john,mary).", and standard window controls. The main area displays the output of the query: "Name: Swethambigai S", "Reg No: 192565064", and "true". A green "1" is visible on the right side of the output. Below the output, there is a query prompt "?- student, father(john,mary)." in a text area. At the bottom of the window, there are buttons for "Examples▲", "History▲", "Solutions▲", a checkbox for "table results", and a "Run!" button.

EXPERIMENT NO. 10

DIETING SYSTEM BASED ON DISEASE

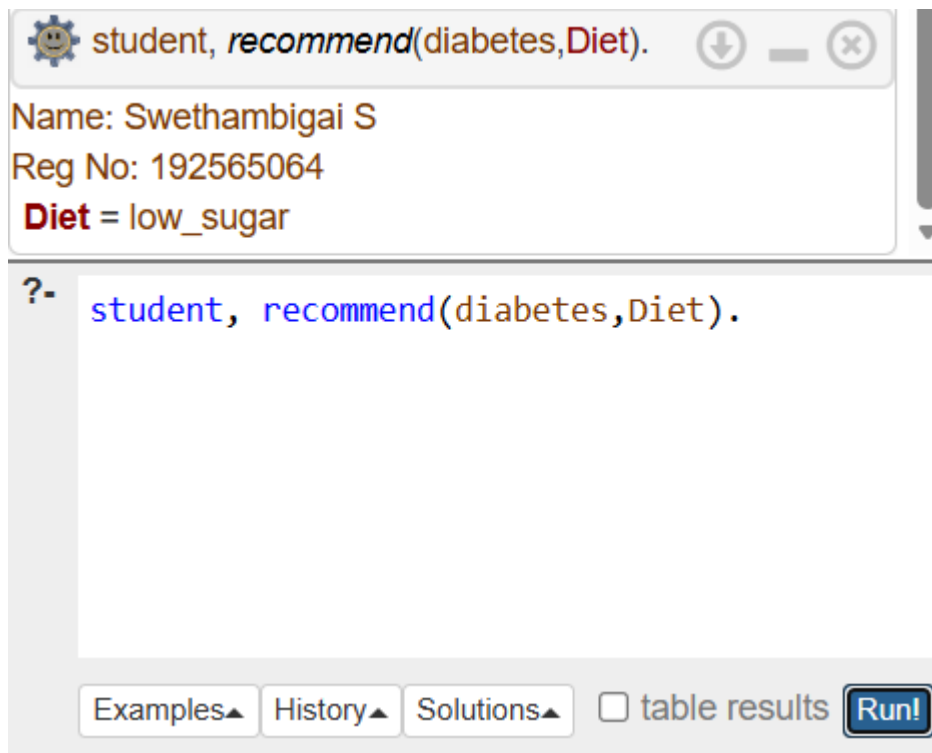
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
diet(diabetes, low_sugar).  
diet(bp, low_salt).  
  
recommend(D, Diet) :-  
    diet(D, Diet).
```

Query

```
student, recommend(diabetes, Diet).
```

Output



The screenshot shows a Prolog IDE window with a title bar containing a gear icon, the text "student, recommend(diabetes, Diet).", and standard window controls. The main area displays the output of the query: "Name: Swethambigai S", "Reg No: 192565064", and "Diet = low_sugar". Below the output, there is a query prompt "?- student, recommend(diabetes, Diet)." in a text area. At the bottom, there are buttons for "Examples", "History", "Solutions", a checkbox for "table results", and a "Run!" button.

```
student, recommend(diabetes, Diet).
```

Name: Swethambigai S
Reg No: 192565064
Diet = low_sugar

?- student, recommend(diabetes, Diet).

Examples History Solutions ☐ table results Run!

EXPERIMENT NO. 11

FORWARD / BACKWARD CHAINING

Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
bird(tweety).  
can_fly(tweety).  
  
forward :-  
    bird(tweety),  
    write('Tweety is a bird').  
  
backward :-  
    can_fly(tweety),  
    write('Tweety can fly').
```

Query

```
student, forward.
```

Output



The screenshot shows a Prolog interpreter window with a title bar containing a gear icon and the text 'student, forward.'. The window has standard minimize, maximize, and close buttons. The main area displays the output of the query: 'Name: Swethambigai S', 'Reg No: 192565064', 'Tweety is a bird', and 'true'. A green '1' is visible in the bottom right corner of the output area. Below the output area, there is a prompt '?-' followed by the query 'student, forward.'.

EXPERIMENT NO. 12

MONKEY BANANA PROBLEM

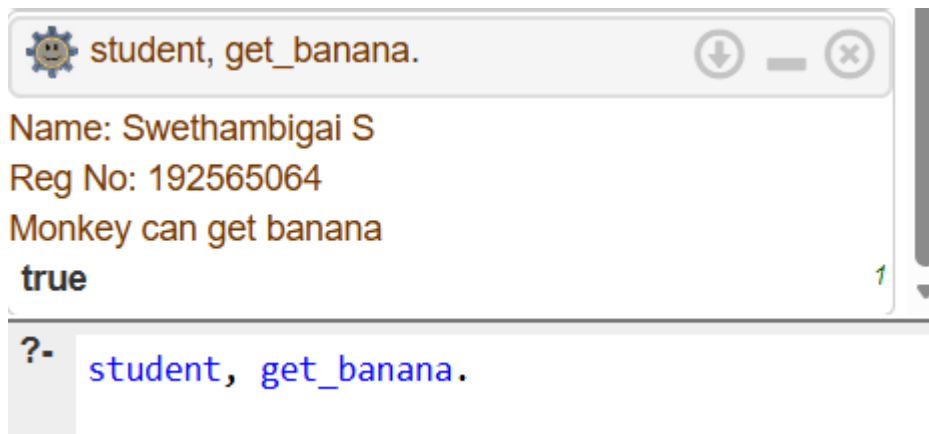
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
monkey_at(door).  
box_at(window).  
banana_at(ceiling).  
  
get_banana :-  
    monkey_at(door),  
    box_at(window),  
    banana_at(ceiling),  
    write('Monkey can get banana').
```

Query

```
student, get_banana.
```

Output



The screenshot shows a Prolog interpreter window with a title bar containing a gear icon and the text 'student, get_banana.'. The window has standard minimize, maximize, and close buttons. The main area displays the output of the query in a monospaced font: 'Name: Swethambigai S', 'Reg No: 192565064', 'Monkey can get banana', and 'true'. A green '1' is visible at the bottom right of the output area. Below the output area, there is a prompt '?-' followed by the query 'student, get_banana.' in a blue font.

```
student, get_banana.  
Name: Swethambigai S  
Reg No: 192565064  
Monkey can get banana  
true  
?- student, get_banana.
```

EXPERIMENT NO. 13

FRUIT AND COLOR – BACKTRACKING

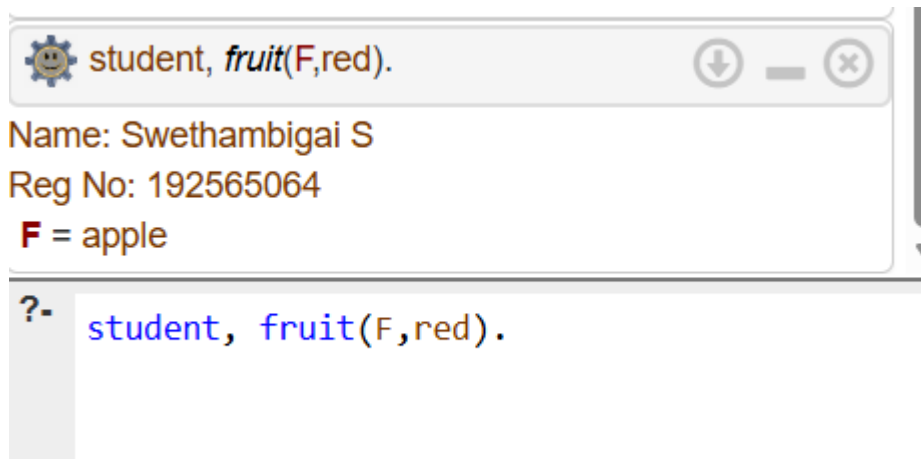
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
fruit(apple, red).  
fruit(banana, yellow).  
fruit(grape, green).
```

Query

```
student, fruit(F, red).
```

Output



The screenshot shows a Prolog interpreter window with a title bar containing a gear icon, the text "student, fruit(F, red).", and standard window controls (down arrow, minus, close). The main area displays the output of the query: "Name: Swethambigai S", "Reg No: 192565064", and "F = apple". Below the output, a prompt "?-" is followed by the query "student, fruit(F, red).".

```
student, fruit(F, red).  
Name: Swethambigai S  
Reg No: 192565064  
F = apple  
?- student, fruit(F, red).
```

EXPERIMENT NO. 14

BIRD FLY OR NOT

Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.
```

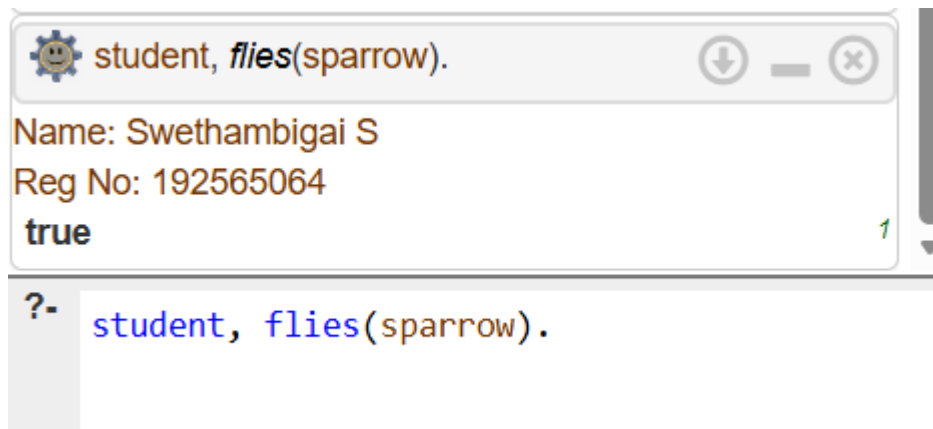
```
bird(sparrow).  
bird(penguin).  
can_fly(sparrow).
```

```
flies(X) :-  
    bird(X), can_fly(X).
```

Query

```
student, flies(sparrow).
```

Output



The screenshot shows a Prolog interpreter window with a title bar containing a gear icon, the text "student, flies(sparrow).", and standard window controls. The main area displays the output of the query: "Name: Swethambigai S", "Reg No: 192565064", and "true". A green "1" is visible in the bottom right corner of the output area. Below the output area, a query prompt "?-" is followed by the text "student, flies(sparrow).".

```
student, flies(sparrow).  
Name: Swethambigai S  
Reg No: 192565064  
true  
?- student, flies(sparrow).
```

EXPERIMENT NO. 15

GRADE FIND

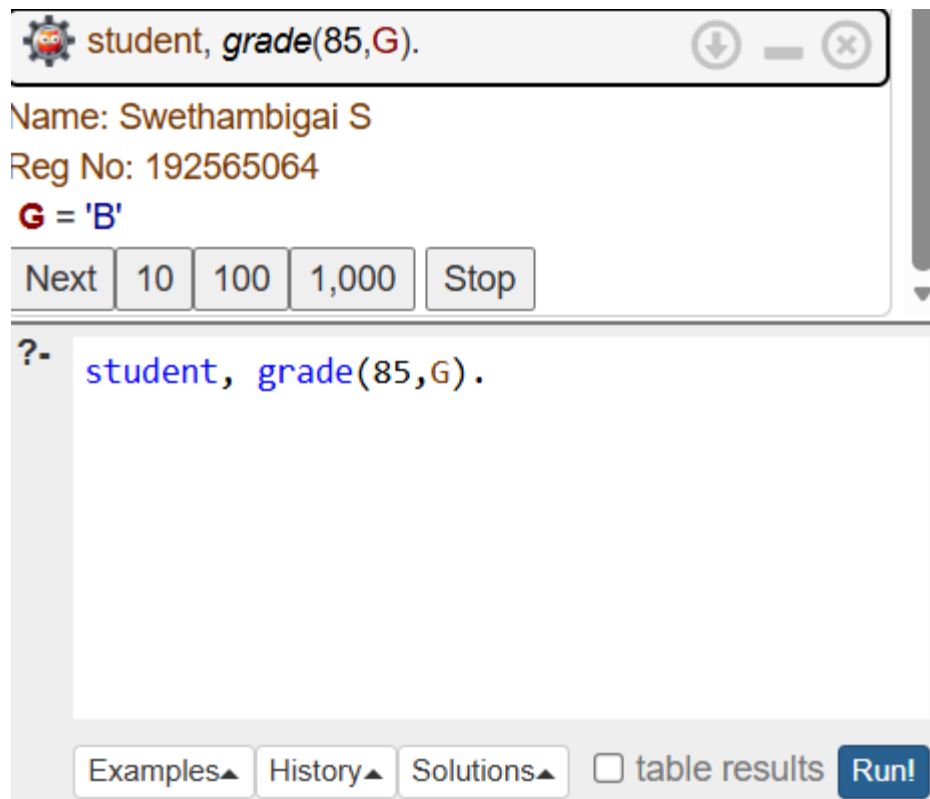
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
grade(M,A) :- M >= 90, A = 'A'.  
grade(M,B) :- M >= 80, M < 90, B = 'B'.  
grade(M,C) :- M >= 70, M < 80, C = 'C'.  
grade(M,D) :- M >= 60, M < 70, D = 'D'.  
grade(M,F) :- M < 60, F = 'F'.
```

Query

```
student, grade(85,G).
```

Output



The screenshot shows a Prolog interpreter window with a title bar containing a gear icon and the text "student, grade(85,G)". The window has standard minimize, maximize, and close buttons. The main area displays the output of the query: "Name: Swethambigai S", "Reg No: 192565064", and "G = 'B'". Below the output, there are buttons for "Next", "10", "100", "1,000", and "Stop". At the bottom of the window, there is a prompt "?- student, grade(85,G)." and a row of buttons: "Examples▲", "History▲", "Solutions▲", a checkbox labeled "table results", and a "Run!" button.

EXPERIMENT NO. 16

EB BILL PROGRAM

Prolog Code

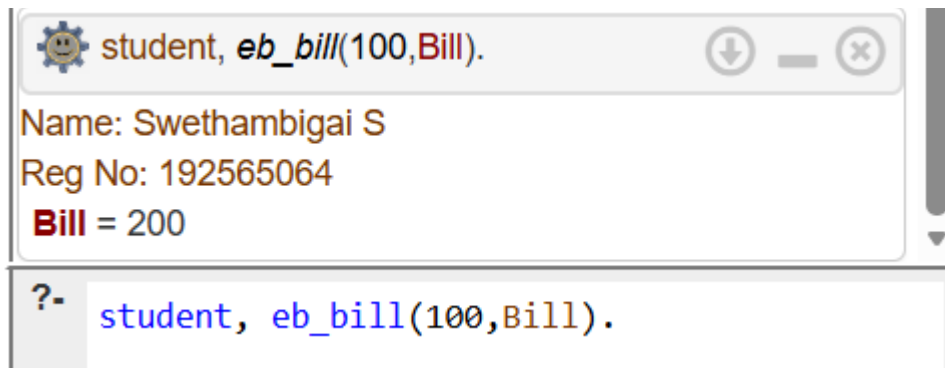
```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.
```

```
eb_bill(Units,Bill) :-  
    Bill is Units * 2.
```

Query

```
student, eb_bill(100,Bill).
```

Output



The screenshot shows a Prolog interpreter window with a title bar containing a gear icon, the text "student, eb_bill(100,Bill).", and standard window controls (down arrow, minus, close). The main area displays the output of the query: "Name: Swethambigai S", "Reg No: 192565064", and "Bill = 200". A vertical scrollbar is on the right. Below the output area, a prompt "?-" is followed by the query "student, eb_bill(100,Bill).".

```
student, eb_bill(100,Bill).  
Name: Swethambigai S  
Reg No: 192565064  
Bill = 200  
?- student, eb_bill(100,Bill).
```


EXPERIMENT NO. 17

STUDENT DATABASE

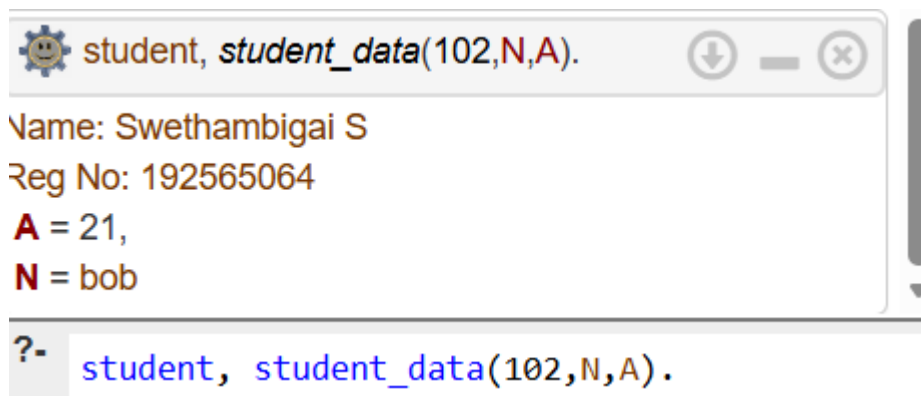
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
student_data(101,alice,20).  
student_data(102,bob,21).  
student_data(103,charlie,19).
```

Query

```
student, student_data(102,N,A).
```

Output



EXPERIMENT NO. 18

PRINT 1 TO 10

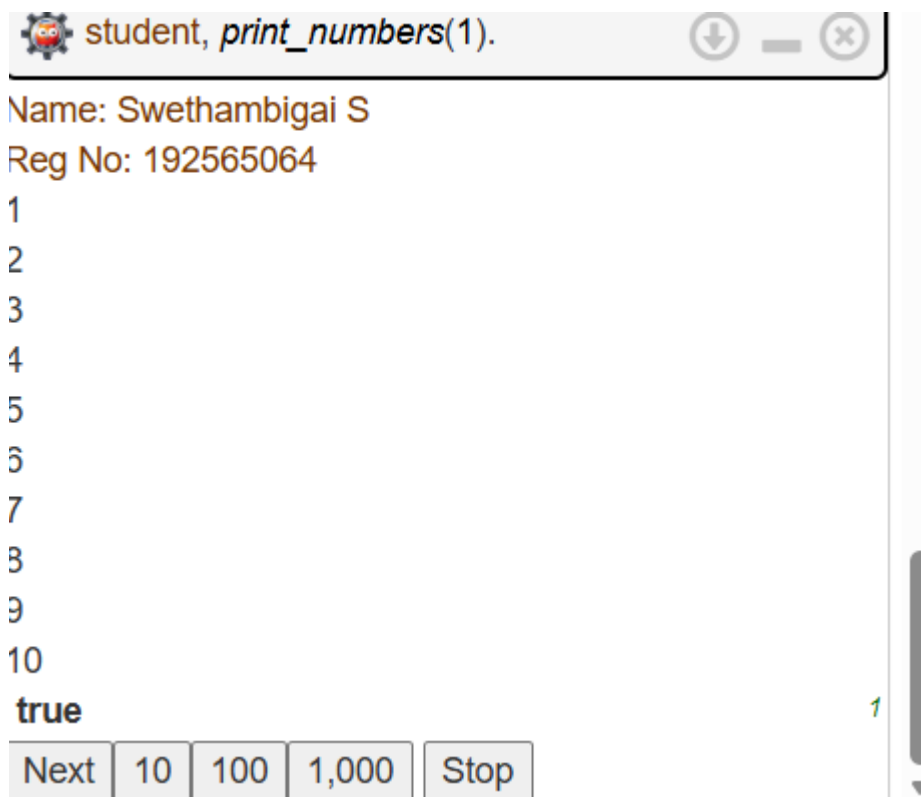
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
print_numbers(11).  
print_numbers(N) :-  
    write(N), nl,  
    N1 is N+1,  
    print_numbers(N1).
```

Query

```
student, print_numbers(1).
```

Output



The screenshot shows a Prolog interpreter window with the title bar "student, print_numbers(1).". The window contains the following output:

```
Name: Swethambigai S  
Reg No: 192565064  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
true
```

At the bottom of the window, there is a row of buttons: "Next", "10", "100", "1,000", and "Stop". A green cursor is visible on the right side of the window.

EXPERIMENT NO. 19

N QUEENS PROBLEM

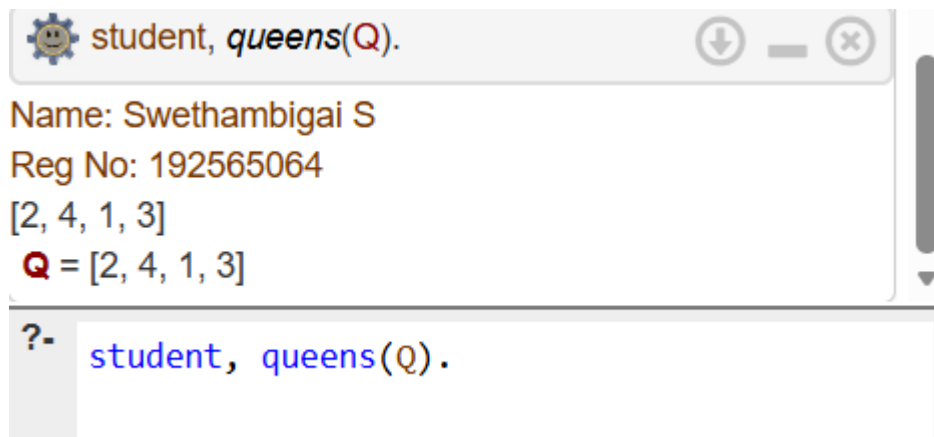
Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
queens(Q) :-  
    Q = [2,4,1,3],  
    write(Q).
```

Query

```
student, queens(Q).
```

Output



The screenshot shows a Prolog interpreter window with a title bar containing a gear icon, the text "student, queens(Q).", and standard window controls (down arrow, minus, close). The main area displays the output of the query: "Name: Swethambigai S", "Reg No: 192565064", "[2, 4, 1, 3]", and "Q = [2, 4, 1, 3]". A vertical scrollbar is on the right. Below the output area, a prompt "?-" is followed by the query "student, queens(Q)." in a monospaced font.

```
student, queens(Q).  
Name: Swethambigai S  
Reg No: 192565064  
[2, 4, 1, 3]  
Q = [2, 4, 1, 3]  
?- student, queens(Q).
```

EXPERIMENT NO. 20

REVERSE A NUMBER

Prolog Code

```
student :-  
    write('Name: Swethambigai S'), nl,  
    write('Reg No: 192565064'), nl.  
  
reverse_number(N,R) :-  
    reverse_number(N,0,R).  
  
reverse_number(0,R,R).  
reverse_number(N,A,R) :-  
    N > 0,  
    D is N mod 10,  
    N1 is N // 10,  
    A1 is A*10+D,  
    reverse_number(N1,A1,R).
```

Query

```
student, reverse_number(1234,R).
```

Output

